

Visualizing Languages as Networks of Meaning

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Abstract. The conventional belief in modern linguistics is that the relationship of sound to meaning in languages is fundamentally arbitrary. This assumption is false and as a result little or no progress has been made uncovering the true nature of languages. When investigated on the merits, meaning in a language can be shown to be part of a holistic system (of meaning) with a precise beginning, a distinctive architecture and a growth trajectory analogous to other living organisms. The difference between a meaning system and its biological counterpart lies in the fact that whereas the latter grows by the division and subdivision of cells, the meaning (or thought) system of a language grows by an iterative process of bifurcating concepts, beginning with the concept ‘One’. Such systems can be represented as three-dimensional networks of interrelated and hierarchically ordered meaning relations that, while dynamic, always retain the same underlying architecture. Using the English language as a model, this paper focuses on how the system emerges, its gross design and how recurrent patterns in words can be used to understand the underlying architecture of the system.

Keywords: Languages as complex systems, meaning and network theory, modeling languages as systems, self-organization of meaning systems, architecture of meaning, arbitrariness of the sign, bifurcation of concepts.

1 Introduction

“When you change the way you look at things, the things you look at change.” – Max Planck

When you think about the parts list of a language, you probably think about its spoken sounds, words, grammar, and syntax; perhaps also the writing system, letters of the alphabet, and so on. In other words you think about the audible and visual dimensions of a language – its outward manifestations. But there is a whole other dimension to language, namely a hidden world of meaning (“Meaning”) often regarded as a proxy for thought.

The goal of my presentation today is to show you that behind what we think of as a language is a complex network of meaning-sound relations, a network that can be represented as a hierarchy of concepts starting with the concept ‘ONE’. The movement required to transition from seeing languages, as languages, to seeing them as meaning systems (“Meaning Systems”) takes a paradigm shift, but if we are to understand the true nature of languages it is vital that we embrace that shift. I will refer to this network of meaning-sound relations in this paper as either a Meaning System, a System or a Network.

A word of caution is appropriate here: Linguists do not buy into the narrative advanced in this paper. The reason is that modern linguistics has built into its credo a dogma that states that there is no relationship between the form of a word and its meaning. In their belief system the sound signal and the orthographic signs that make up the word 'onion' are regarded as unrelated to the thing 'onion'. It is merely a convention that we call it that. It could have been called a 'house', a 'turnip' or anything else for that matter.

If that were true, then there would be no structure to Meaning within languages worth investigating. The entire domain would be an unstructured set, about as interesting as a pile of stones. As you will have already have surmised, I believe that this conclusion is wrongheaded, empirically falsifiable and just plain wrong. I'll leave it to you to decide on the merits where you stand.

The first step on the road to recognizing that languages are more than the sum of their parts, and more than just a means of communication, is to learn to see them as Meaning Systems. Section 2 of the paper looks at how the study of the forgotten/lost domain of Meaning in the English language leads to a fundamental realignment of our understanding of what a language is.

Section 3 reviews the architecture of the System as a whole. It investigates on a theoretical level why the concept ONE and the idea of One-ness represents the first concept in a Meaning System. I also discuss the key architectural design feature of all languages, a feature that I have called polarity of meaning (or "POM" for short). POM refers to the structural organization of concepts in a Meaning System into logical pairings of opposites (or complements). Section 4 speaks to the nature of One-ness and why the concept is represented in the way it is in terms of both sound and symbol.

Section 5 addresses the idea that there are fields of Meaning within the overall architecture of the System. These fields are associated with each of the phonetic sounds that exist within a given Meaning System. I conclude with a review of some of the major features of the System and share some thoughts about the implications of the unearthing of this hidden system of meaning-sound relations that underlies all languages.

Although the specific examples that I reference throughout my talk are drawn from the English language, I want to make it clear that my thesis is that the general architecture of languages discussed in Section 3 applies to all languages, living and dead.

2 Languages are Meaning Systems

A language is considered by the general public to be synonymous with its external features. But this is a superficial way to view a language because it overlooks the entire substructure of Meaning that underlies every language. This is the equivalent of mistaking a tree for that portion of the tree that appears above ground and ignoring everything else that lies beneath the surface.

What is important in life is invariably hidden: Life begins in the womb; the germination of plants happens below the ground; and the vital energy that animates living things is invisible to the eye. And what is true of biological organisms is also true of languages. The life breath of a language lies beneath its surface forms. If your goal is to understand the essential nature of a language, you cannot focus solely on its external features to the exclusion of the unseen dimension of Meaning.

Of course, these two levels of a language – its substructure and its superstructure – are not independent of each other. Meaning expresses itself throughout the entire structure of a language. And just as the roots of a tree absorb water and nutrients from the soil, and form a structural system that supports the trunk and crown of the tree, so the substructure of a language supports the externalized and visible superstructure of a language.

While the discipline of Linguistics does study Meaning, it does so at the level of sentences and you have to go deeper than that if you want to understand the nature of the Meaning System underlying a language. Sentences clearly represent a later stage in the developmental growth of a language. Before there were sentences there had to be words capable of being strung together into sentences. My contention is that you have to look into Meaning at the level of words in order to see what lies beneath the surface forms.

My claim is that, contrary to the accepted wisdom in the field of linguistics, there is a relationship between the sound sequence of a word and its meaning. In other words, the sound sequence 'o-n-i-o-n' is related to the thing 'onion'(1). It is not simply an arbitrary relation born of convention. Moreover, the procedure that maps sound to meaning in the unconscious happens in a structured way and in line with a set of prescribed rules. As a consequence, the DNA of words in terms of Meaning is embedded within the structure of the words themselves. And this entire process is self-organizing. There is no puppeteer behind the scenes pulling the strings.

What you are seeing when you look at a word is the final expression of a System that starts with a generative procedure that maps sound to meaning (or meaning to sound) in the unconscious. If a word can be regarded as the fruit of the tree of Meaning, then its ingredients are formulated deep in the roots of the Meaning System in line with a simple set of rules.

To facilitate the leap from seeing languages, as languages, to seeing them as integrated systems of meaning, it is useful to highlight the differences between these two perspectives. That is the purpose of the accompanying Figure 1 which highlights the paradigm shift that takes place when you integrate Meaning into the equation of how we look at a language.

LANGUAGES	MEANING SYSTEMS
External to us – object of scientific inquiry	Internal – regarded as beyond scientific enquiry
Communication system	System of thought
Signs arbitrary – conventional, have to be learnt	Signs not arbitrary – form related to meaning
Names reference things	Names are descriptions of things
Invented	Biological endowment
Conscious	Unconscious
Linear – strings of sounds	Hierarchical
Origins speculative	Origins are traceable by study of network connections
6500 or so examples	One meaning system, each employing different set of phonemes
Well Studied	Ignored

Fig. 1. Languages vs. Meaning Systems

Viewing a language as a network allows us to ask questions that were previously closed off to inquiry. Imagine for a moment a three-dimensional branching network of concepts in which the nodes are concepts and the edges represent the links between concepts. In such a Network we could investigate the order of appearance of concepts – the older, and therefore earlier concepts, lying closer to the center of the network, and the newer ones being further from the center. We could also inquire into the growth of the Network by identifying internal markers associated with the different stages of its development.

These kinds of inquiries into the hidden domain of a language are not without precedent in the realm of scientific inquiry. By learning how to interpret the evidence found in the growth rings of a tree, for example, you can learn valuable information about both the age of the tree and the prevailing climatic conditions at each stage of its growth. Similarly, ice cores provide information about climatic conditions of a given locale over spans of tens of thousands of years. According to the website ice.cores.org, ice cores ‘are essentially frozen time capsules that allow scientists to reconstruct climate far into the past.’

In the same way, studying the underlying substructure of Meaning allows us to reconstruct the emergence and growth of languages. We can ask, for example, what concepts emerged first in the Network. We can also ask what developmental stages a language goes through on its way to maturity. The principles employed in these kinds of inquiry relative to languages are of the same nature as the inquiries into the growth rings of trees and of ice cores. They investigate a hidden domain by having recourse to data manifest in the internal structure of the object under investigation.

Accessing the System

But how can we go about studying a System that is hidden and not accessible to consciousness? The answer is actually deceptively simple. Rather than looking at a word as an arbitrary string of sounds (or orthographic signs in the written word), whose meaning is given by convention, we need to look at a word as potentially encoding Meaning directly into its spoken or written form.

When we look at words from this perspective, we can begin to investigate how Meaning is built into words and the rules that guide the process. Of course, you cannot expect to extrapolate any rules about how the underlying System works, or what form the architecture takes, from the study of an individual word. A word contains only a fractional part of the coding embedded in the System as a whole.

The analogy with a jigsaw puzzle is useful in this regard. You cannot look at one piece of a 500-piece jigsaw and project what the final image of the finished puzzle will be. However, each individual piece of the jigsaw contains within its distinctive shape and color pattern, a fragment of the coding that you need to build up a representation of the final image. The key is to fit the pieces together to reveal the completed image.

Likewise, each word in the lexicon of a language contains within it a fragment of the coding of the underlying Meaning System such that we can, at least in theory, reconstruct the architecture of the System by studying the coding of sound-meaning relations in words and recognizing the relationship between the interlocking parts. Approached in this way, each item of decoded information identified in words adds to the picture of architecture of the System as a whole.

Where the analogy of a jigsaw puzzle and a Meaning System breaks down of course is in the sheer magnitude and complexity of the latter. A Meaning System is a vast, three-dimensional Network with hundreds of thousands of words and therefore coding fragments. To give you an idea of the scale of the Network, consider that the Second Edition of the 20-volume Oxford English Dictionary (OED), published in 1989, contains full entries for over 170,000 words in current use in English.

When you adopt this methodological approach, you find that there are patterns in the way Meaning is built into words *and* into the architecture of the Meaning System as a whole. These patterns are clearly not random because they follow a distinct logic and are consistently applied. In the aggregate, these patterns reveal important information about the architecture and nature of the System.

The traditional paradigm is blind to these patterns (and the underlying information carried in them), due to a preconceived belief that there is no connection between the form of a word and its meaning. As a consequence, when these patterns are pointed out, they are either ignored or explained away as coincidence.

3 The Origins and Architecture of a Meaning System

I indicated in the introduction to this paper that a Meaning System can be represented as a hierarchy of concepts starting with the concept 'One'. I want to take minute to explain, by way of a thought experiment, why it is that the concept ONE necessarily represents the beginning of the network of concepts and the starting point of any Meaning System.

Take a look around you wherever you happen to be and pay attention to everything you see and hear. As you look around, be aware that your perceptual field is made up of separately identifiable features in your local environment. If you are indoors you might see walls, windows, doors, a floor, a ceiling, chairs, a ceiling fan, and other individuals or animals. In the world outside, you have the sky, the ground, cars, houses, trees, buildings, and so on. And these individuated, and apparently separate items, are made up of smaller pieces – people have heads, bodies, legs and arms; buildings have windows, floors, roofs; trees have trunks, limbs, and leaves, etc. Each of these units has a name.

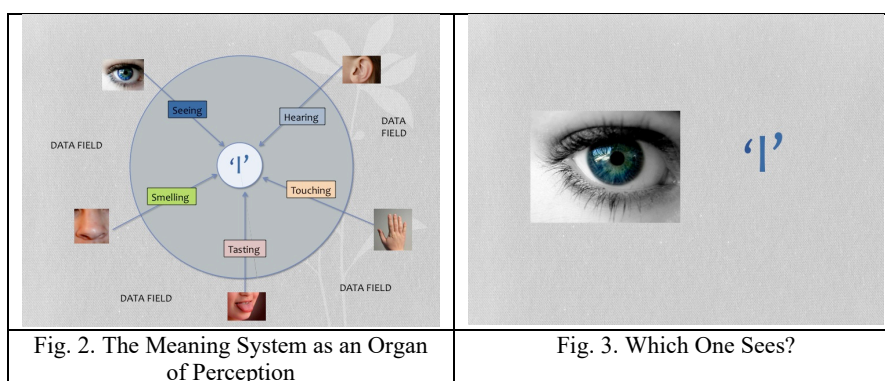
We take it for granted that the separation of elements in the overall gestalt has always been the bedrock of our perceptual reality. However, as Soviet psychologist Lev Vygotsky (1896-1934) demonstrated in a series of experiments, conducted in the 1920s, there is strong evidence to suggest that newborn infants begin with an awareness of an undifferentiated whole and only later graduate to being able to pick out individual objects from the background.

How then do we make the transition from seeing the undifferentiated whole to seeing the world as made up of a fabric of separate entities? The answer I think lies in the acquisition of a language, or as we are now reframing it a Meaning System. It's the installation of a Meaning System – or more accurately, the exercise of this biological endowment – that causes a separation to occur between you (as you) and the rest of the world. At the same time, the separation sharpens the differentiation of all things out there in the mind external world that are identified as not being you.

I am sure this interpretation of the facts will come as something of surprise to those of you that take for granted that it is your senses that are responsible for the distinctions I am attributing to Meaning Systems. After all, the argument runs, you can see with your own eyes that these objects out there in the world

are separate from each other and from you and, in the case of sounds, you can hear with your own ears the distinct and separate quality of that experience. It seems patently obvious.

But the fact is your senses don't have the ability to make such interpretative distinctions. It is not the physical eyes that allow you to see anything. They are merely receptors that process light. Seeing is a percept of the mind/brain. You 'see' with the mind/brain, in other words with the inner 'I', shown in Figure 2. The same is true of all the other senses. Hearing, touching, tasting and smelling are all percepts of the mind/brain.



This inner 'I', 'installed' when you were a toddler, functions to synthesize and interpret the data coming in from the senses, organizing it into an intelligible vision of the world. And it does so on the basis of naming according to the distinguishing feature of the entity being named. And this inner 'I' is a construct of language and of the Meaning System to which it belongs. In light of this information, consider the question posed in Figure 3 above.

It is the Meaning System, operating as an organ of perception, that allows us to separate things out one from another. The reason this is important in the context of what I'm saying about languages and Meaning Systems is that you can't name anything until you are able to separate out entities from the undifferentiated background. Before you can imbue an entity with a nametag or label you have to perceive it as an individuated 'One' within an integrated perceptual reality that includes other Ones.

The entities listed in the thought experiment conducted above were basically all of one kind. They were all entities that take up either two- or three-dimensional space depending on the nature of the object described. All of these things are Ones in the terms of our discussion because they are perceived as entities separate and apart from everything else in your perceptual field. But there is another category of 'One' not explicitly mentioned in the category of items

listed in the thought experiment. These are entities that take up temporal space, metaphorically speaking. This would include any action that has a beginning and an end – my lifting my arm from point A to point B, for example. Sounds also qualify under this category because they have a beginning, a crescendo and an ending. This category also includes periods of time. All of these temporal events are perceived as a unique and individuated ‘Ones’ and are integrated into the field of our perceptual reality with the spacial Ones.

Before a tree is a tree, it’s a ‘One’; a baseboard is a ‘One’; a car is a ‘One’; the sky is a ‘One’; the blast of a fog horn is a ‘One’; the time that elapses between the rising and setting of the sun is a ‘One’; and so on. But to be able to see something as a ONE – remember you are seeing with your inner linguistic ‘I’ – you must first have an internalized concept of ONE-ness that can be mapped onto the world, and you cannot attain that awareness until you are sufficiently differentiated as an entity to be able to hold onto the concept of ONE-ness.

The thesis that I am propounding is that a Meaning System necessarily starts with the concept ‘One’. The origins of the System are now hidden from view because all things – both spacial and temporal – seem to be so different qualitatively, and have diverse names, but historically and perceptually they were all Ones before they were given names. And, as we have seen, that grounding still underlies our perception of the world as seen through the lens of a Meaning System. The function of names is to distinguish one ‘One’ from another ‘One’ on qualitative grounds.

Just as an infant has to learn to separate things out from the undifferentiated whole, so we can be fairly certain on an intuitive level at least that early Man would have had to cross that same threshold in order to progress from being a pre-linguistic creature to being able to express him- or herself in symbolic language. You need an inner ‘I’ to make that leap.

4 The Nature and Representation of ‘One-ness’

In the introduction to this paper, I alluded to the fact, albeit not immediately transparent, that there is a connection between the form of a word and its conceptual meaning. At that time, I emphasized that the key to unlocking that connection is simply to bracket the idea that there is no such connection in favor of an empirical approach based on evidence. If we are to accord any level of credence to this idea then there should be a discoverable correlation between the sound and the symbol for the concept ONE and its meaning in the English language. As it turns out, there is indeed a very profound correlation between the two.

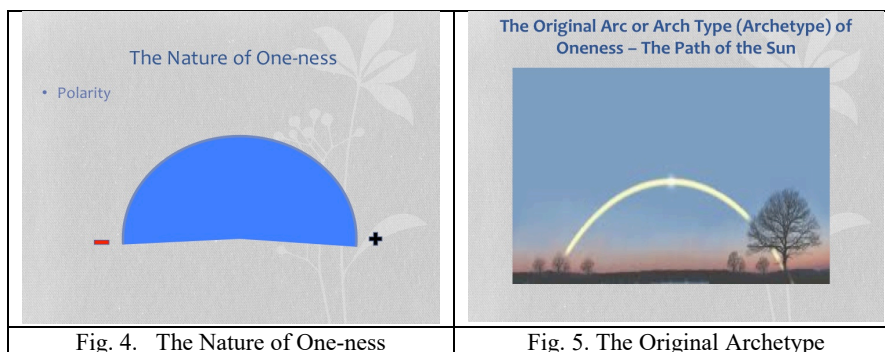
In order to understand the correlation between form and Meaning, as it pertains to One-ness, you have to ask the following question: What is the distinguishing feature of One-ness, something inherent in all Ones, that could be the basis of

the naming of the concept ONE? In other words, what do all Ones have in common?

The answer is that they exhibit polarity, or otherwise stated, complementarity. All spacial entities (remember we are calling them ‘Ones’), depending on whether they are two- or three-dimensional objects, will have an up and a down, a top and bottom, a back and front, and inside and an outside, a left and right, etc. All temporal entities – also Ones – will have a beginning and an end, a before and after, an earlier and a later, and so on. This fact is universally true. Oneness is defined by polarity. If something doesn’t have polarity embedded in it, it can’t be a ‘One’ because it has no existence, either spacially or temporally, at least none that we can detect.

How can we represent Oneness in symbolic terms? The answer is shown in Figure 4 below. The arc in this figure beautifully captures the symmetry of One-ness because it incorporates polarity (or complementarity) and the transition between one state and the other through the form of the arc. It also represents the spacial or temporal unity of the individuated ONE in the colored area that separates the ONE from the surrounding background.

Where does this symbol come from? I would speculate that it comes straight from nature in the form of the daily path of the sun through the sky, as shown in Fig. 5 below. This would have been a phenomenon whose daily transitions early Man would have been intimately familiar, and one he would no doubt register on the cave wall with a scratch mark to record the passing days.



The properties of this symbol are incorporated directly into both the lower and upper case forms of the letter N, as illustrated in Figure 6 below. N is the orthographic symbol that represents the concept ONE in a great many languages, including English, German (eiN/eiNe), Dutch (eeN), Norwegian (eN), Swedish (eN), Danish (eN) French (uN, uNe), Italian (uNo), and many others.

In terms of the Meaning Systems underlying these languages, the key thing to note is that the symbol stands for concepts expressing both unity and separation:

1. The sound /n/ and the symbol 'n' (and 'N') convey the idea of Oneness.
2. The sound /n/ and the symbol 'n' (and 'N') convey the idea of its opposite, not One-ness or No.

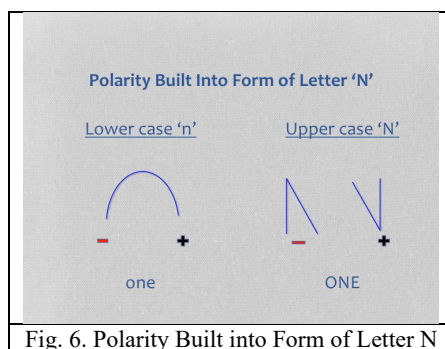


Fig. 6. Polarity Built into Form of Letter N

The important point to note is that the Meaning System encompasses both meanings. N can either act to integrate Two into One (here the shared diagonal bar in the upper case letter serves as a connecting line) **and** to separate One into Two (here the shared bar serves as a dividing line).

When standing for the concept of One-ness, N is added to the end of a word. This lets you know that the entity described is perceived as an independent unit, separate and apart from the fabric of the overall perceptual field into which it is integrated. Here are some examples of N endings performing that functional role:

- n = an elementary particle: anion, antimuon, boson, carbon, cation, codon, electron, fermion, gluon, ion (an atom carrying a positive or negative charge), meson, neutrino, neutron, operon, photon, positron, etc. (*Only the consonants in these words carry Meaning.*)
- The modifier added to the root of the word gives you the type of 'One' being named. Ask what kind of 'One' are we talking about?
Answer: **Phot-on = Light One.**
- Sun, moon, ocean, nation, station, fountain, mountain, brain, etc. In these examples, the type of One described can be understood only in relation to the fields of Meaning that precede and modify the root meaning. Thus the SuN is a particular type of One whose properties are defined by the field of S.

When standing as a marker of negation, the N comes at the beginning of the word and modifies a concept. Thus Dependent becomes iN-Dependent, Consequential becomes iN-Consequential, Divisible becomes iN-Divisible and Conscious becomes uN-Conscious. Note that the N, in this position, can take any concept and split it into two. You simply take the selected concept and place on the right side of the arc. Then draw an arc from the right pole to its end point on the left. The concept then represented at the left pole bears a meaning that stands in opposition to (or is a complement to) the selected concept on the righthand side of the arc.

Our understanding of the concept of ‘Oneness’ and its counterpart ‘Not One’, and the relationship between the two, allows us to tentatively draw out the beginning of the Meaning System or Network as it relates to the English language. See Figure 7 below. The orthographic symbol ‘N’ that sits in the center of the map of Meaning highlights the fact that two basic meanings inhere in one symbol (and by extension one sound), namely One-ness (to the left) and its opposite – No or ‘not Oneness’ (to the right).

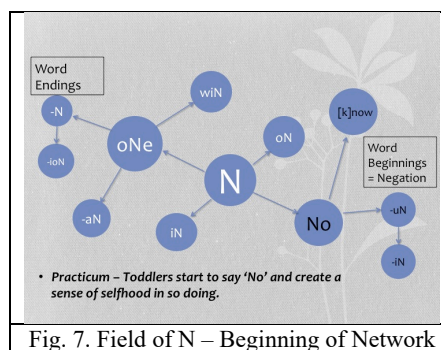


Fig. 7. Field of N – Beginning of Network

As a sidebar, it is interesting to note that there is a connection between toddlers ability to say ‘No’ and the development of their formation of the notion of One-ness. A toddler is in effect saying, ‘I am not any of these Ones out there in the world. I am Me, a separate being or ‘One’.

Polarity of Meaning is the architectural design feature that defines a Meaning System. It is embedded in the concept for One-ness at the inception of the Network and extends throughout the System and throughout its life. The point to be made here is that polarity turns out to be everywhere in languages. It is a self-organizing design feature of all languages.

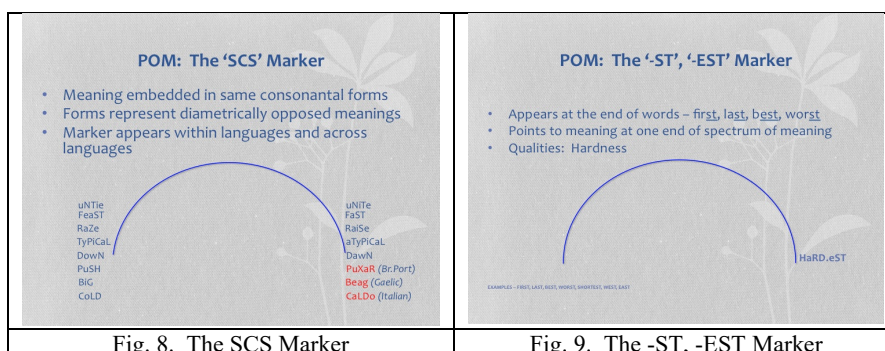
If you think about it for a moment, you will understand why POM is embedded into the architecture of any Meaning System. The explanation is that concepts come in pairs. You cannot have one pole in a spectrum without the presence of

the other. There is no such thing ‘left’ without ‘right’, ‘up’ without ‘down’, ‘inside’ without ‘outside’, ‘life’ without ‘death’, etc. One pole implies the other and grounds it. When one concept is in the foreground, the other is in the background. When one is conscious, the other is unconscious. It is therefore a logical necessity that they enter into a Meaning System at the same time.

Although negation is by far the most prevalent marker of POM in a Meaning System, it is not the only one. The figures below illustrate two other markers of polarity within the Meaning System as expressed in the English language. The first, shown in Figure 8, illustrates words with the same consonantal structure (the SCS Marker) but having polar opposite meanings. Interestingly, the examples of this particular marker of polarity exist within a single Meaning System, as for example the words ‘FaST’ and ‘FeaST’, but also exist across languages. An example of the latter phenomenon is the word ‘BiG’ in English and its counterpart ‘BeaG’ in Gaelic which means ‘small’.

It stretches credulity to imagine that this type of patterning is purely coincidental. A much more rational explanation for the pervasive presence of this design feature in and across Meaning Systems is that POM is present in the architecture of all such systems, and that the trees of Meaning represented by such Networks display parallel developmental growth.

The second marker shown below in Figure 9 (the -ST or -EST Marker) has to do with words which end in either ‘-st’ or ‘-est’. The inherent Meaning of this root word is as an indicator that what is being pointed to is the most extreme incarnation of that particular quality or property on the spectrum, in this case HARDNESS. The other pole represents the least hard incarnation of the property of HARDNESS.



5 Fields of Meaning

I have already alluded to the fact that within the domain of Meaning there is an association of particular sounds with particular concepts. Although there are

only twenty-six letters in the alphabet, twenty-one of which are consonants and therefore the carriers of Meaning, there are approximately 44 sounds (or phonemes) in English, with some variation premised on accent and articulation. Every language contains its own suite of active phonemes. Conceptually, you can think of each of these fields as being distinct branches of the Meaning System or Network.

The field of N, as the first sound-meaning field to come into existence, naturally reflects a pre-naming stage of language development. As previously detailed, everything is a ONE before it is anything else. As a result, Meaning in this field is most often associated with abstract ideas. However, there is a small universe of mind external objects that are named within the field, including lumps, hill-like structures, circles, simple marine organisms, sharp objects and worms. Body parts – such as nose, nape, neck, noggin, [k]nee, navel, nail [k]nuckle and navel are also included. This is consistent with the idea that the field represents an early developmental stage of the English language.

Each field of sound-meaning relations probably unfolds in a predetermined order – yet to be determined – just as the branches of a tree represent bifurcations of earlier branches or limbs. And like the number system, it appears to be the case that the naming system incorporates place value into its overall architectural design, such that the meaning of a sound will change according to its placement in a word. When you add these kind of calculations into the mix of sound-meaning relations, it is easy to see why Meaning Systems are as complex as they are and why too they appear to be so opaque from the outside looking in.

The field of M appears to represent a later stage of growth in the development of a Network, because it embodies concepts having to do with relationships between existing Ones. The symbol 'm' is made up of two n's joined together ($n+n = m$) and reflects the field of M's embodiment, in one of its guises, as the relationship component (the interaction) between two Ones. In this regard M can be interpreted as corresponding to the number 3 in the number system, the letter symbol being the same as the number symbol but rotated anticlockwise through 90 degrees.

6 Conclusion

When seen through the lens of a Meaning System, we can see that languages are a great deal more than the sum of their parts and more than simply a means of communication. They are organs of perception that determine how we see the world. In particular, our proclivity to perceive polarity in the outside world is actually a projection of the attributes of the Meaning System acting as a filter on our perception, in much the same way as a red filter on a camera superimposes something on an image that is not actually there. The reason that this aspect of

languages is not part of our everyday consciousness is that we have no memory of our reality prior to the activation of the Meaning System.

Another conclusion that we can draw from the set of facts described above is that the number system and the naming system are very likely parallel systems. The number system starts with the number one. The naming system starts with the concept One. Both these systems share the unique property of discrete infinity, a rare property that permits the generation an infinite number of expressions from a discrete number of elements.

The implications of making a paradigm shift from seeing languages, as languages, to seeing them as Meaning Systems are dramatic both in terms of opening up a more expansive understanding of the true nature of languages, and for potentially unraveling some of the some of the greatest mysteries of our time – including the Hard Problem of Consciousness, the Origin of Languages and How Information is stored in the brain.

It is to be hoped that the lessons learned from developing a broader understanding of the nature of languages will help spawn new research programs designed to flesh out the inner life and architecture of languages and in so doing pave the way forward into these uncharted territories of the human mind.

References

- (1) The form of the word ‘oNioN’ incorporates two Ns. One of the meanings assigned to this form of consonantal structure is a circle or cycle, as in the words, aNNual, aNNular, aNNiversary, and NiNe (units 1 -9). If you now imagine cutting an onion in half, you will understand why the word oNioN takes the form it does.